Nikita Singh

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Summary

AI/ML Engineer specializing in Generative AI, LLM systems, explainable AI, and computer vision pipelines. IEEE Best Paper awardee with experience building agentic AI platforms, legal NLP systems, and deployable ML workflows across clinical and enterprise applications. Backed by research publications, benchmark-driven evaluation, and full-stack AI product development.

Skills

- **Languages:** Python, C++, TypeScript, JavaScript, SQL
- **Machine Learning / AI:** PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, Hugging Face, NLP, Deep Learning, U-Net, SHAP, LIME, MediaPipe, OpenCV
- **Generative AI / LLMs:** Claude API, Groq API, Llama 3, LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM Agents
- **Backend and APIs:** Next.js, React, Node.js, Tailwind CSS, Supabase, PostgreSQL, REST APIs
- **Data and Tools:** DuckDB, Pandas, Streamlit, Plotly, Docker, Git, MySQL, Vercel, AWS

Experience

Sun Pharmaceutical Industries Ltd. May 2025 – Jul 2025
ML Research Intern
- Built ensemble classification models for ADR prediction (Decision Tree, Random Forest, Voting Classifier) on 10K+ clinical records; improved prediction accuracy by **12%** against baseline ensemble performance.
- Applied SHAP and LIME attribution across **50+ clinical variables**, enabling interpretable ADR prediction and clinical decision support.
- Built an ML pipeline for 10K+ clinical records, supporting **5+ models** and reducing experimentation time by **30%**.

EnviroVision Aug 2025 – May 2026
AI Engineer Intern
- Architected AI-powered smart waste segregation system using CNN-based vision models trained on a **22K-image dataset** with IoT sensor streams for real-time waste classification.
- Deployed deep learning pipelines achieving **97.6% precision** for real-time edge inference.
- Engineered sensor-fusion pipelines processing 100+ sensor readings per sec using ultrasonic, inductive, and vision sensors.

Projects

- **Nexus — AI Freelance Escrow Platform** [GitHub] [Demo]
  LLM workflow using Llama-3.3-70B via Groq auto-decomposes project briefs into milestones and triggers conditional micropayments. Implemented adaptive reputation scoring with payout/refund logic and secured multi-tenant authentication using Supabase PostgreSQL RLS. *Stack: Next.js, TypeScript, Groq, Supabase*
- **Aurelius AI — Explainable Legal Intelligence Platform** [GitHub] [Demo] [IEEE]
  Deployable legal intelligence platform for contract analysis, document summarization, and explainable case insights across uploaded legal documents. Implemented entity extraction, confidence scoring, and citation-aware retrieval workflows across 6 legal domains. *Stack: React, TypeScript, Vite, SHAP, LIME, Legal NLP*
- **RosterIQ — Agentic Analytics System** [GitHub] [Demo]
  3-tier memory architecture (Episodic, Procedural, Semantic) for natural-language SQL over DuckDB. Self-improving agentic AI system via Claude Tool Use API with auto-generated Plotly dashboards. *Stack: Claude API, Streamlit, DuckDB, Pandas*

Education

Chandigarh University Mohali, India
Bachelor of Engineering(Hons.) — Computer Science and Engineering (AI and ML) | CGPA: 8.7/10 *Aug 2022 – Jun 2026*

Publications

- **Legal Document Analysis Using Deep NLP** [IEEE Xplore]
  Transformer-based Legal-BERT framework integrating rhetorical role labeling, abstractive summarization, and SHAP/LIME explainability; improved legal summary coherence while reducing review effort by **60%**.
- **DAL-GAN-PS: Dual-Adversarial Linguistic GAN** [IEEE Xplore]
  Dual-adversarial GAN with PhonoScript constraints for low-resource multilingual text generation; achieved **42.67 BLEU**, **91.2%** morphological accuracy, and **88.9%** Sandhi compliance across 8,742 multilingual sentence pairs.
- **Visual Trajectory Fields in Deep Vision Models** [IEEE Xplore]
  XAI framework analyzing representation trajectories across ResNet, ViT, and CLIP architectures under adversarial and illusion-based inputs using trajectory divergence, curvature, and instability metrics.

Key Achievements

- **1st Place** — Snap Syntax Full Stack Hackathon, IIT Roorkee Cognizance 2026 [GitHub] [Certificate]
- **3rd Place** — Spectral Bridge ML Hackathon, IIT Roorkee 2026 [GitHub] [Certificate]
- **Best Paper Award** — IEEE iCONECCT-2025 for DAL-GAN-PS research in adversarial linguistic generation [Certificate]
- **MICCAI LiTS Benchmark** — Achieved Dice Score of **0.91** using U-Net-based liver tumor segmentation

Certifications

- **Oracle** — Cloud Infrastructure Generative AI Professional (1Z0-1127-24) [Certificate]
- **IBM** — Predictive Analytics (Advanced) [Certificate], Security Governance and Law [Certificate]
- **University of Michigan** — Python for Everybody Specialization [Certificate]
- **Microsoft** — Generative AI for Data Analysis [Certificate]